

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2018-2019 (Even Semester)

Innovative Practices Description

UNIT I MOS Transistor Principle

Degree, Semester & Branch: VI Semester B.E ECE A

Course Code & Title: EC6601 VLSI design

Name of the Faculty member: Mrs.R.Chitra

Name of the Topic: IC Fabrication

Name of the Innovative Practice: Animation Video

Description:

An integrated circuit, or IC, is small chip that can function as an amplifier, oscillator, timer, microprocessor, or even computer memory. An IC is a small wafer, usually made of silicon, that can hold anywhere from hundreds to millions of transistors, resistors, and capacitors. These extremely small electronics can perform calculations and store data using either digital or analog technology. The steps I have outlined below are the most basic stages in the fabrication process, and many steps are repeated, altered, or omitted – depending on chip design. The fabrication steps are as follow as:

1. **Start with sand:** The process of creating a computer chip begins with a type of sand called silica sand, which is comprised of silicon dioxide. Silicon is the base material for semiconductor manufacturing and must be pure before it can be used in the manufacturing process.
2. **Silicon ingot:** A purified silicon ingot, which weighs around 100kg, is shaped from melted silica and made ready for the next step.
3. **Cut wafers:** The circular silicon ingot is sliced into wafers as thin as possible while maintaining the material's ability to be used in the fabrication process.
4. **Photolithography:** After being polished and readied for the process, a layer of photoresist is spread thinly across the wafer. This layer is then exposed to a UV light mask, which is shaped in the pattern of the microprocessor's circuits.
5. **Ions and Doping:** Exposed photoresist is washed off and the silicon wafer is bombarded with ions in order to alter its conductive properties – this is called doping.
6. **Etching:** A pattern of hard material is applied to the wafer using another photolithography step. Chemicals are then used to remove unwanted silicon, leaving behind thin silicon ridges.
7. **Layering Interconnects:** All the transistors are now connected in an architecture which allows the chip to function like a processor. The layering and design of these interconnects is incredibly complex, and there can be over 30 layers of metal connections in a single processor.

8. **Packaging:** Dies are packaged with a substrate and heat spreader, and assume the familiar form factor of a desktop processor. The heat spreader conducts heat away from the silicon and into the heatsink mounted on top of it. Processors are then tested for power efficiency, maximum frequency, and other performance metrics.

All these fabrication steps are explained to the students through animated video to clearly comprehend the concept visually.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	-	-	-	-	-	1	-	2	1	-

CO1: The student will be able to explain the basic characteristics of MOSFET and different CMOS process technologies

References:

1. <https://mybroadband.co.za/news/hardware/200748-how-a-computer-chip-is-created-from-sand-to-cpu.html>
2. <https://www.youtube.com/watch?v=Q5paWn7bFg4>